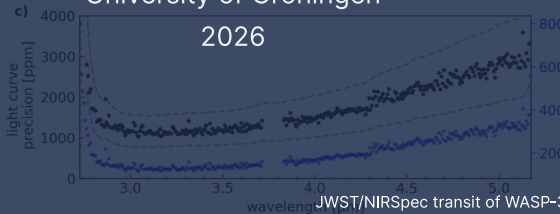
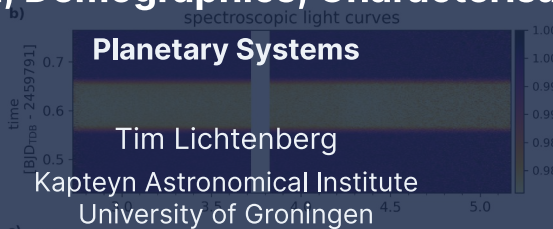
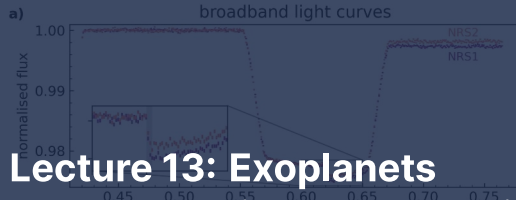


Lecture 13: Exoplanets

Detection, Demographics, Characterisation



Learning Objectives

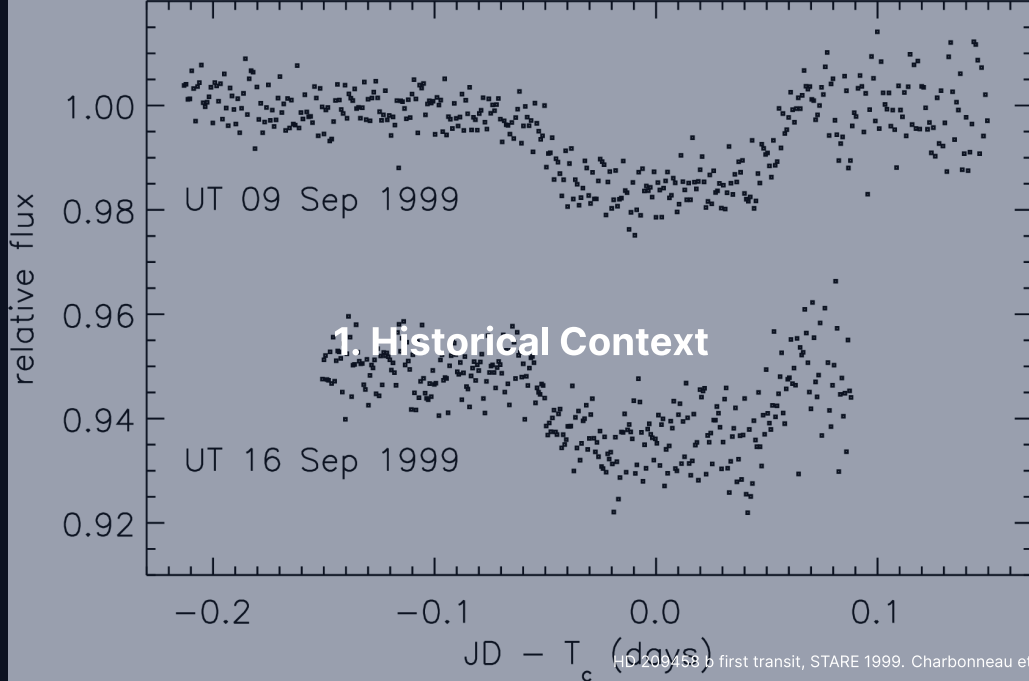
By the end of this lecture, you will be able to:

- Describe the main exoplanet detection methods and their biases.
- Interpret the period-radius diagram and the radius valley.
- Apply transit + radial-velocity geometry to derive mass, radius, and bulk density.
- Evaluate JWST-era atmospheric results and biosignature claims.

From zero confirmed exoplanets in 1991 to more than 6000 in 2026.

Recap: Where We Are in the Course

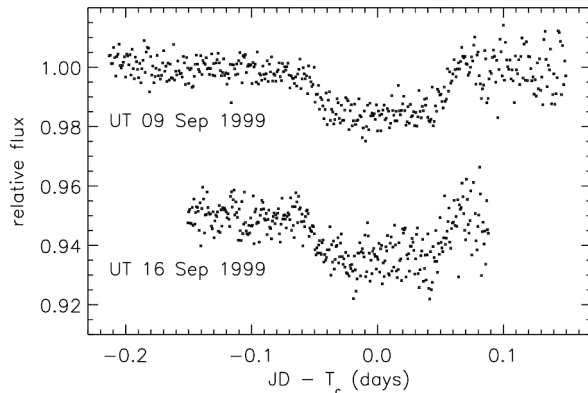
- L9-L11 walked through every major solar system planet.
- L12: small bodies as formation fossils.
- Today: *turn the question outward*. Are these patterns universal?
- L14: synthesis. What constitutes a convincing detection of life?



From Zero to Six Thousand in Three Decades

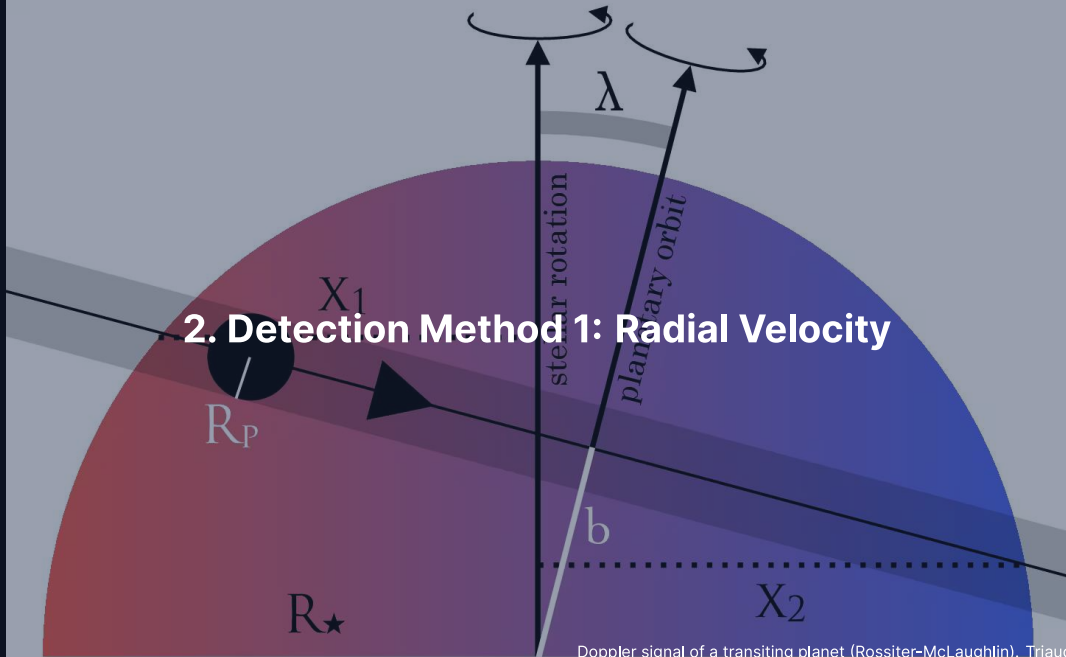
- **1992:** Wolszczan + Frail. Two planets around pulsar PSR B1257+12 from pulse-timing.
- **1995:** Mayor + Queloz. **51 Pegasi b** via radial velocity. $0.5 M_J$, 4.23-day orbit.
 - Contradicted every textbook picture of giant-planet formation.
 - → migration models reshaped formation theory.
 - Nobel Prize 2019.
- **1999:** HD 209458 b. First transit of a known RV planet.
 - Combined transit depth + RV → first bulk density of an exoplanet.
 - Confirmed hot Jupiters are gas-dominated.
- **2009–2018:** Kepler + K2 transformed the field statistically.
- **2022–present:** JWST atmospheric characterisation era.

HD 209458 b: First Ground-Based Transit (1999)



- STARE photometer, two nights in Sep 1999.
- Successive transits dropped flux by $\sim 1.7\%$.
- Independent detection by Henry et al. (2000).
- After this point: exoplanets were physical objects, not just Doppler signals.

2. Detection Method 1: Radial Velocity



Radial Velocity: Doppler Reflex of the Star

- Star + planet orbit a common barycentre.
- Star's reflex orbit: $a_{\star} = (m_p / M_{\star}) a_p$.
- Spectral lines blueshift / redshift periodically.
- For circular, edge-on orbit:

$$K_{\star} = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{m_p \sin i}{M_{\star}^{2/3}} \frac{1}{\sqrt{1 - e^2}}$$

Quick scales:

- Jupiter at 5.2 AU: $K_{\star} \approx 12.5 \text{ m s}^{-1}$.
- Earth at 1 AU: $K_{\star} \approx 9 \text{ cm s}^{-1}$.

Instrument Floor and Stellar Noise

- ELODIE (1995): $\sim 10 \text{ m s}^{-1}$.
- HARPS (2003): $\sim 1 \text{ m s}^{-1}$.
- ESPRESSO (2018): 10 cm s^{-1} demonstrated.
- Now **stellar noise**-limited: granulation, p-modes, starspots produce 10 cm s^{-1} to 1 m s^{-1} jitter even on quiet stars.
- True Earth twin: at the very edge of physical accessibility.

The $m \sin i$ Degeneracy

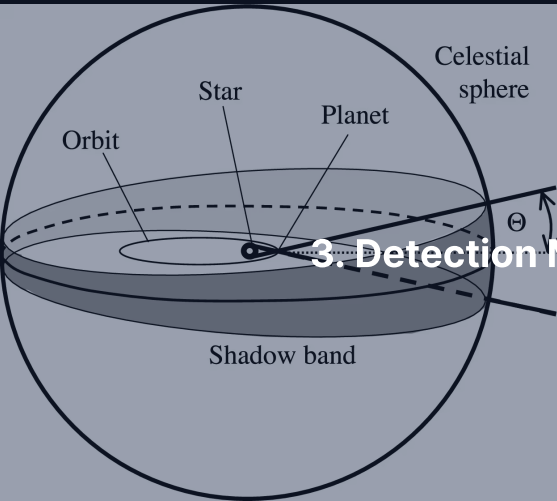
RV gives only K_* , hence $m_p \sin i$.

- Face-on: $\sin i = 0$, no signal at all.
- Edge-on: $\sin i = 1$, true mass.
- RV alone $\rightarrow$ *minimum* mass only.

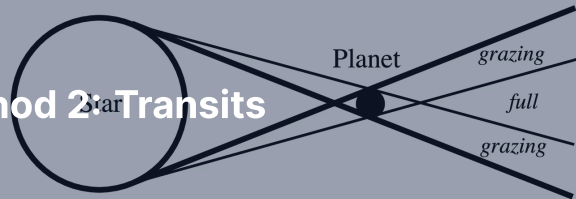
Break the degeneracy with:

- **Transit** ($\sin i \approx 1$ to a few percent),
- **Astrometry** (gives inclination directly),
- **Direct imaging** (geometric).

RV biases: massive planets, short orbits, bright slowly-rotating quiet G/K stars.

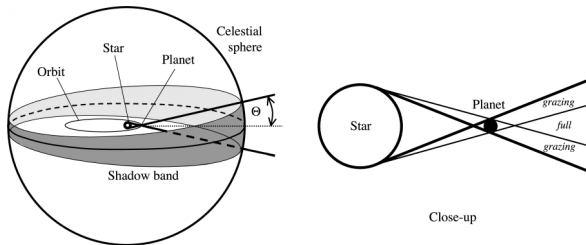


3. Detection Method 2: Transits



Close-up

Transit Geometry



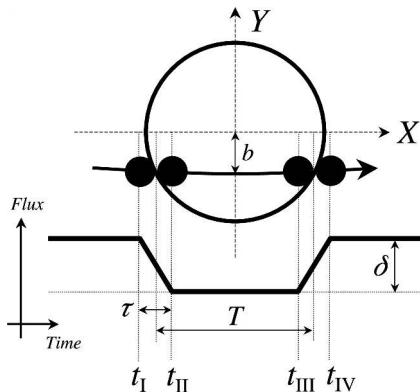
- Orbital plane near our line of sight $\rightarrow$ planet crosses stellar disk.
- Depth: $\delta = (R_p/R_\star)^2$.
- Probability: $\Theta \approx R_\star/a$.
- Strong selection: short periods + large $R_p/R_\star$.

Quick Scales

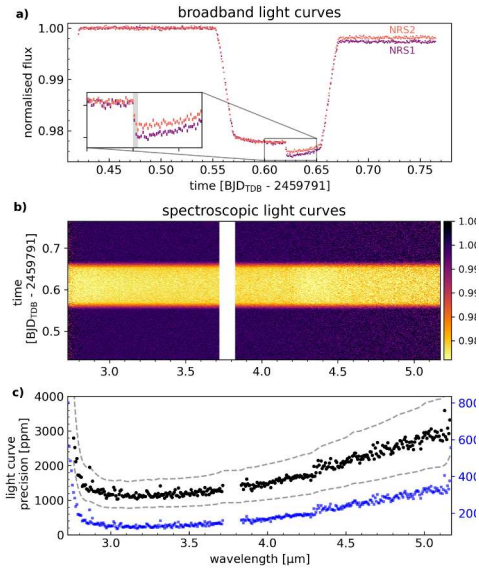
- Jupiter (R_J) around Sun ($R_\odot$): $\delta \approx 1.01\%$. Easy from the ground.
- Earth around Sun: $\delta \approx 84$ ppm. Requires space.
- Earth around M dwarf ($R_\star = 0.15 R_\odot$): $\delta \approx 4000$ ppm. Ground-accessible!

Inverse scaling with stellar radius is why TRAPPIST-1 dominates atmospheric characterisation.

Light Curve Anatomy



- Four contact times: t_I , t_{II} , t_{III} , t_{IV} .
- Duration $\rightarrow$ impact parameter b , inclination.
- Ingress shape $\rightarrow$ stellar limb darkening.
- Depth $\rightarrow$ planet radius (given R_*).

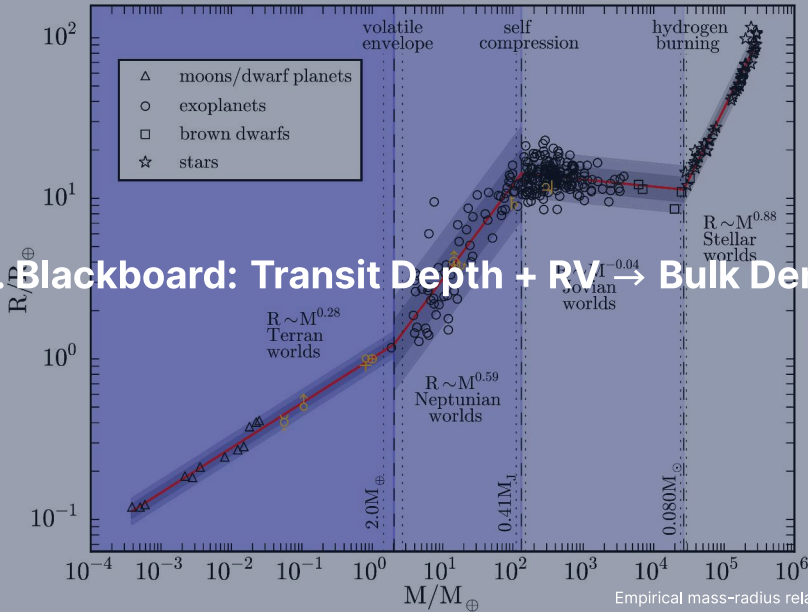


JWST/NIRSpec transit photometry of WASP-39 b: 200–500 ppm precision per channel.
Alderson et al. (2023).

Reading the JWST Transit Photometry

- **Top:** raw broadband transit light curves, NIRSpec G395H NRS1 + NRS2.
- **Middle:** spectroscopic light curves at each wavelength bin.
- **Bottom:** per-channel precision $\rightarrow$ 200–500 ppm per integration.
- Photometric noise floor is now *below* typical exoplanet absorption features.
- Selection biases:
 - Short period ($P < 100$ days) for repeated transits.
 - Large $R_p/R_\star$ (favours hot Jupiters and M-dwarf systems).

4. Blackboard: Transit Depth + RV $\rightarrow$ Bulk Density



Why Combine Transit and RV?

- Transit alone: R_p , but no mass.
- RV alone: $m_p \sin i$, but no radius.
- Combined: **both** for the same planet → bulk density.
- Density is the central observable that turns a detection into a physical object.

Step 1: Transit Depth

Star = uniform disk R_* . Planet = opaque disk R_p .

$$\frac{\Delta F}{F} = \frac{\pi R_p^2}{\pi R_*^2} = \left(\frac{R_p}{R_*} \right)^2$$

Given R_* from spectroscopy + parallax, the depth gives R_p directly.

Limb darkening + ingress shape $\rightarrow \sim 10\%$ corrections.

Step 2: RV Semi-Amplitude

Conservation of momentum: $M_{\star} a_{\star} = m_p a_p$, $a = a_{\star} + a_p$.

$$v_{\star} = \frac{2\pi a_{\star}}{P} = \frac{m_p}{M_{\star} + m_p} \cdot \frac{2\pi a}{P}$$

Use Kepler: $a^3 = G(M_{\star} + m_p)P^2 / (4\pi^2)$, sub in:

$$K_{\star} = v_{\star} \sin i = \left(\frac{2\pi G}{P} \right)^{1/3} \frac{m_p \sin i}{M_{\star}^{2/3}} \frac{1}{\sqrt{1 - e^2}}$$

$K_{\star} \propto P^{-1/3}$: short-period planets give larger reflex.

Step 3: Combine

A transit forces $\sin i \approx 1$ (orbit near edge-on).

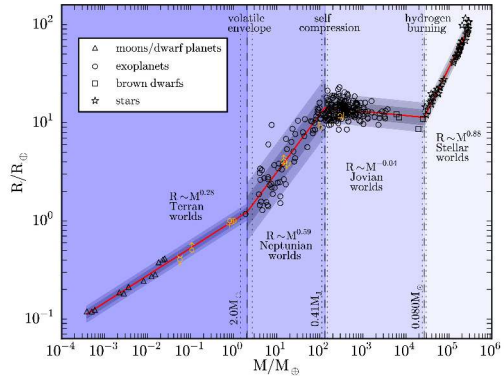
- $\sin i \approx 1$ collapses the $m_p \sin i$ degeneracy.
- Both R_p and m_p measured for the same object.
- Bulk density follows from elementary geometry:

$$\bar{\rho}_p = \frac{m_p}{(4/3)\pi R_p^3} = \frac{3m_p}{4\pi R_p^3}$$

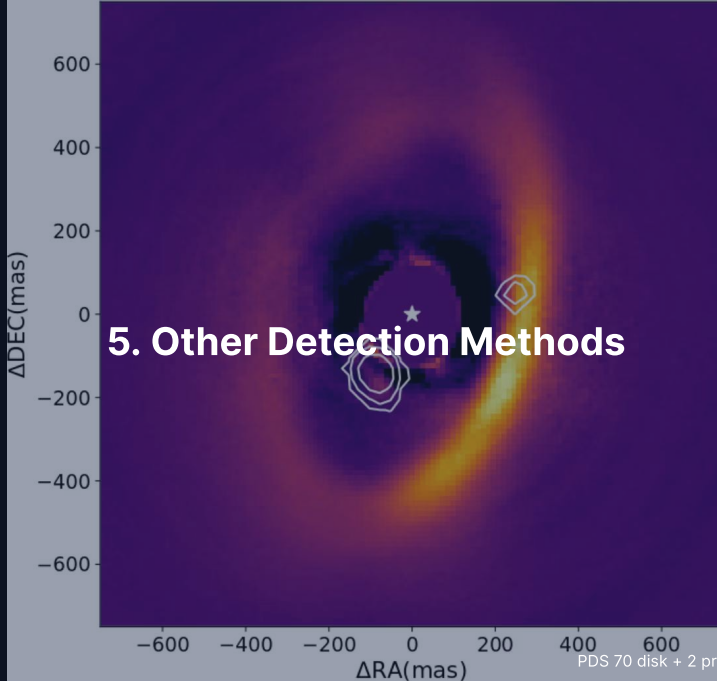
Transit + RV is the single observational machinery that took exoplanet science from exotic claim to compositional census.

Density Maps to Composition

- $\bar{\rho} \approx 5.5$: rocky (Earth, Venus).
- $\bar{\rho} \approx 1.3$: gas giant (Jupiter).
- $\bar{\rho} \approx 0.5$: inflated H/He envelope.
- $\bar{\rho} \approx 2\text{--}4$: *sub-Neptune*, the most common type with no solar-system analogue.



5. Other Detection Methods

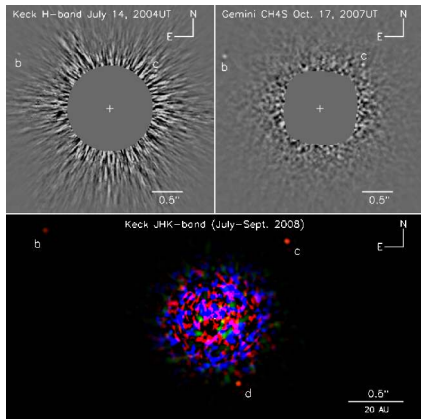


PDS 70 disk + 2 protoplanets. Haffert et al. (2019)

Direct Imaging: Spatial Separation of Photons

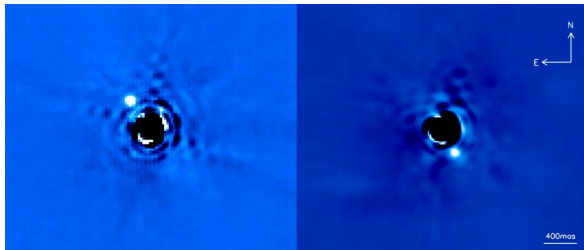
- Conceptually simplest, technically hardest.
- Sun + Jupiter contrast at visible: 10^{-9} . Sun + Earth: 10^{-10} .
- Four ingredients:
 - Adaptive optics (atmospheric turbulence correction)
 - Coronagraphs (stellar light suppression)
 - Angular + spectral differential imaging (speckle subtraction)
 - Long integration ($\rightarrow$ averaging speckle realisations)
- Strongly biased to *young*, massive, wide-orbit, self-luminous giants.

HR 8799: Four Imaged Giants

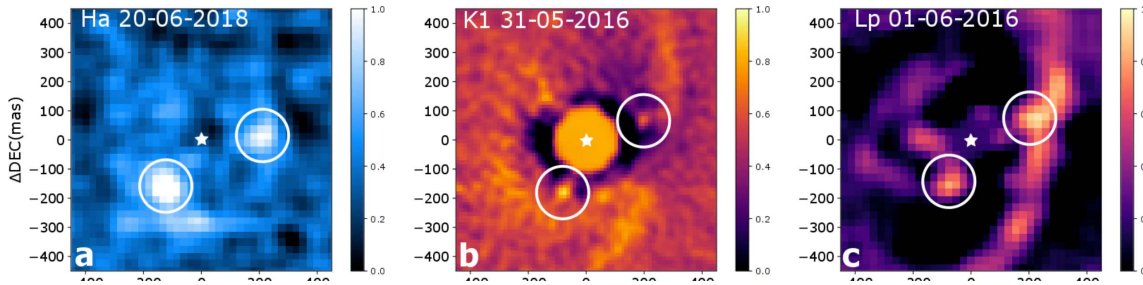


- Marois et al. 2008. Keck + Gemini AO; sub-panels show three discovery epochs (HK 2004, CH₄ 2007, JKH 2008) confirming bound co-moving companions.
- Three planets initially, fourth added 2010.
- Each $\sim 5\text{--}10 M_J$ at 14–70 AU.
- Host: 30 Myr A-type. Planets still self-luminous and cooling.

β Pictoris b: Orbital Motion in Two Epochs

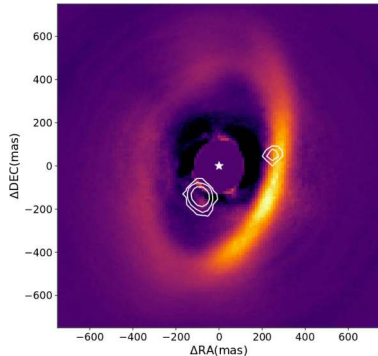


- Lagrange et al. (2010).
- $\sim 8 M_J$ at ~ 9 AU.
- Two epochs (2003, 2009) $\rightarrow$ orbital motion confirms bound companion.
- Same star: imaged debris disk + a second giant (c).



H α confirmation of PDS 70 b + c forming inside a disk gap. Haffert et al. (2019).

PDS 70: Forming Planets in a Disk Gap



- Host: 5 Myr K7 with cleared central cavity.
- SPHERE (NIR) + MUSE ($H\alpha$) detected b inside the gap (Keppler+ 2018).
- Second protoplanet c added later (Haffert+ 2019).
- Both are *actively accreting* ($H\alpha$ shock emission).
- Cleanest direct match between planet-disk theory and observation.

Astrometry: Gaia Will Find Jupiter Analogues

$$\alpha = \frac{m_p}{M_\star} \cdot \frac{a_p}{d}$$

- Jupiter analogue at 10 pc: ~ 0.5 mas. Earth analogue: $\sim 0.3 \mu\text{as}$.
- Hipparcos (1989–93): ~ 1 mas, no exoplanets.
- Gaia (2013–): $20\text{--}50 \mu\text{as}$ per epoch, $\sim 10 \mu\text{as}$ final.
- DR4 (late 2026): $10^2\text{--}10^3$ astrometric exoplanets, mostly Jupiter analogues.
- Astrometry gives *inclination directly* $\rightarrow$ true mass.

Microlensing + Timing

- **Microlensing:** foreground star magnifies background source; planet adds a small spike.
 - Sensitive at kpc distances; sweet spot at 0.5–10 AU.
 - One-shot, never repeats.
 - Roman (2027): expected ~ 1000–3000 bound + comparable rogue planets.
- **TTVs:** gravitational coupling between transiting planets perturbs transit times → dynamical masses without RV.
 - Essential for faint stars: TRAPPIST-1 masses come almost entirely from TTVs.
- **Pulsar timing:** extraordinarily sensitive but tiny target sample.
- **Eclipsing binaries:** Kepler-16 b ("Tatooine") is circumbinary.

Detection Bias Summary

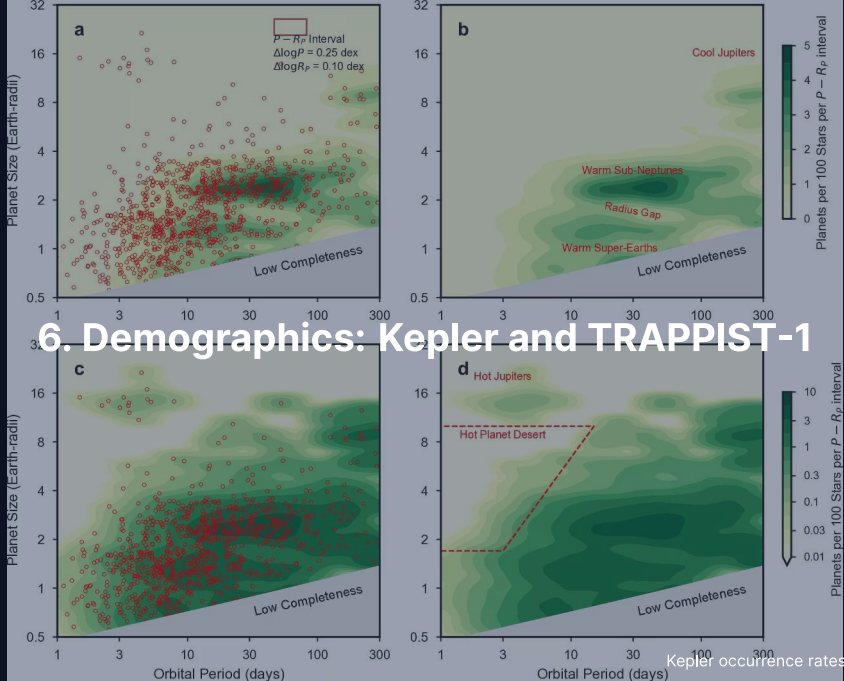
- **RV**: massive, 1–10 yr orbits, bright G/K stars.
- **Transit**: short period, large R_p/R_* , quiet bright stars.
- **Direct imaging**: wide (> 10 AU), young, massive, self-luminous.
- **Astrometry**: wide orbits matching mission baseline.
- **Microlensing**: 1–10 AU, any host distance, unrepeatable.
- **Timing**: compact multis (TTVs) or circumbinaries.

The shape of the observed exoplanet archive reflects the union of biases as much as the underlying physics.

Break

End of Part 1: Detection Methods

Part 2: Demographics, Architectures, Atmospheres

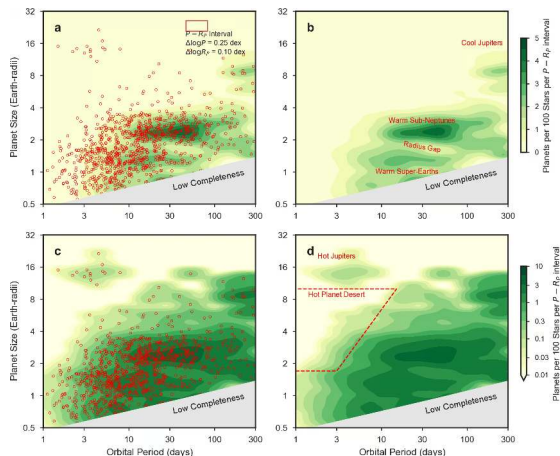


6. Demographics: Kepler and TRAPPIST-1

The Kepler Revolution

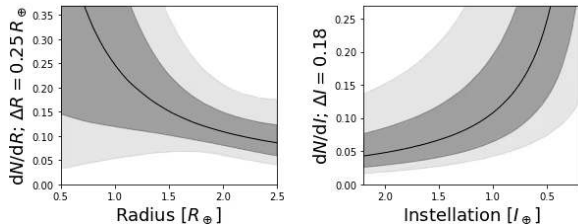
- 0.95 m telescope, single field, 150,000 stars, 4 years, 30 ppm precision.
- *Computable detection efficiency* → first occurrence-rate measurements.
- Headline: at least one planet per main-sequence star (Petigura+ 2018).
- Small planets ($R < 4 R_{\oplus}$) dominate by far.
- Hot Jupiters: 0.5–1% of Sun-like stars (rare, but easy to find).
- $\eta_{\oplus}$ (Earth-size in HZ): ~ 0.1 – 0.6 , central value ~ 0.3 (Bryson+ 2021).

Period-Size Occurrence Rate



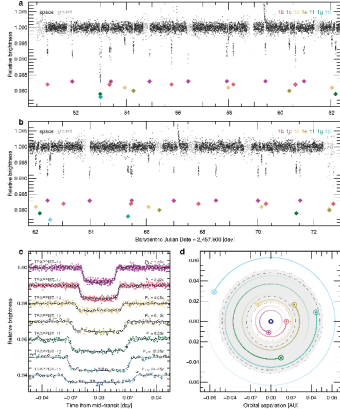
- California-Kepler Survey, after stellar parameter refinement (sub-panels label distinct period-radius regimes).
- Small planets common at every period.
- Slight fall-off at longer periods (partly observational).
- Long-period flattening → Kepler had only 4-yr baseline.

$\eta_{\oplus}$: Earths in Habitable Zones



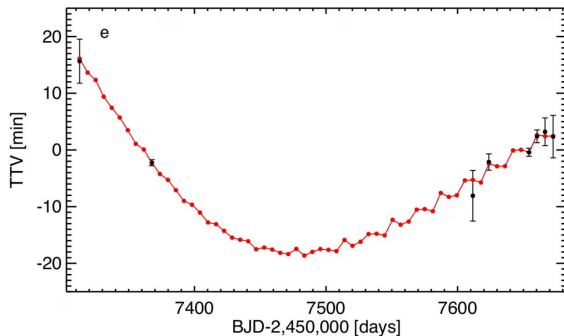
- Bryson+ 2021 marginal occurrence.
- Left: per unit radius. Right: per unit instellation.
- Dark/light bands: 68% / 95% credible intervals.
- $\eta_{\oplus}$: 0.1–0.6 depending on definition.
- **Earth-class planets are ordinary, not miraculous.**

TRAPPIST-1: Seven Earths in Resonance



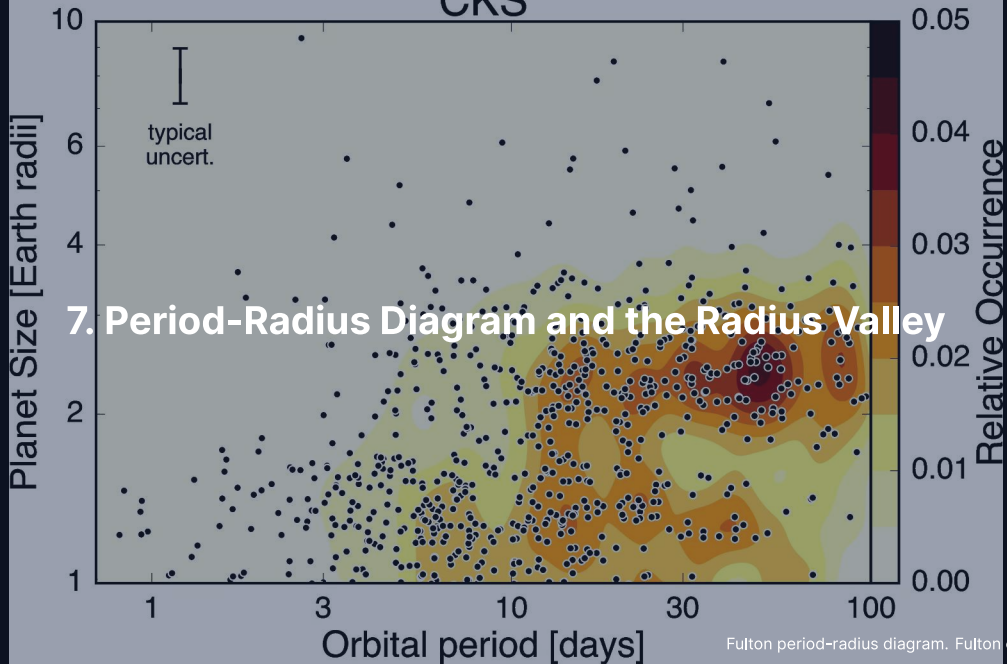
- M8 ultra-cool dwarf at 12 pc.
- Seven transiting Earth-sized planets within 0.06 AU.
- Five-step mean-motion resonant chain → early disk migration.
- Probability all seven transit if random: $< 10^{-3}$.
- Gillon et al. 2017.

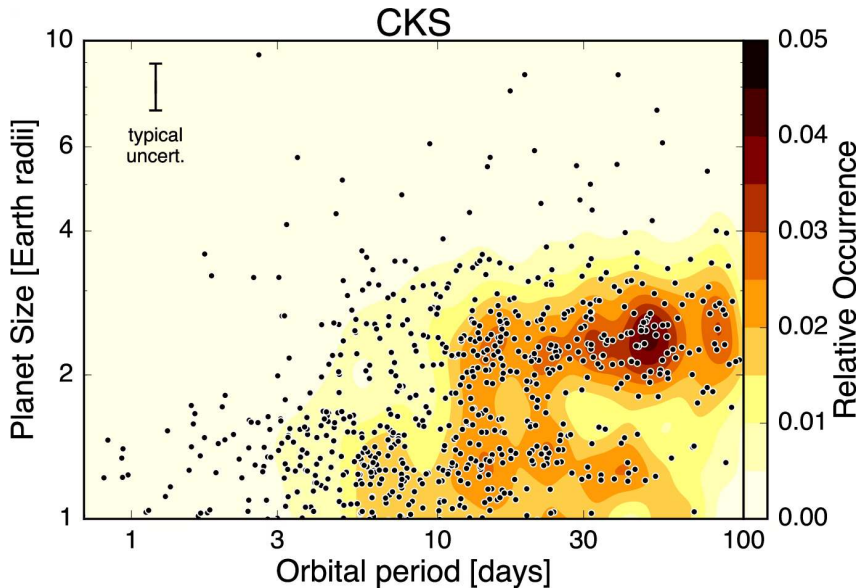
TRAPPIST-1 Masses from TTVs



- TRAPPIST-1 e transit-timing variations.
- Tens-of-minutes amplitude over hundreds of days.
- Best-fit dynamical model includes the full 7-planet coupling.
- Yields all seven masses without RV.
- M8 is too faint for high-precision RV.

CKS



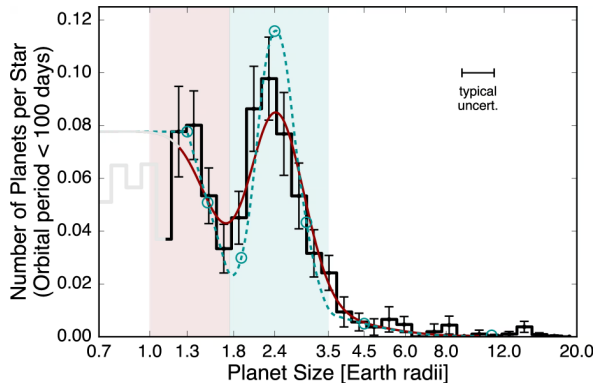


Period-radius distribution of small Kepler planets after stellar parameter refinement. Fulton et al. (2017).

Reading the Period-Radius Diagram

- Hot Jupiters: tight clump at $P < 10$ d, $R > 10 R_{\oplus}$.
- Sub-Neptunes ($2\text{--}4 R_{\oplus}$): *the most common class*; no solar-system analogue.
- Super-Earths ($1\text{--}1.8 R_{\oplus}$): rocky planets larger than Earth.
- Terrestrial regime ($P > 100$ d, $R < 1.5 R_{\oplus}$): largely unexplored. PLATO will fill.
- Most striking feature: the **deficit at** $\sim 1.8 R_{\oplus}$.

The Radius Valley (Fulton Gap)



- Statistically significant deficit at $1.5\text{--}2 R_{\oplus}$.
- Below: super-Earths peaked at $\sim 1.3 R_{\oplus}$.
- Above: sub-Neptunes peaked at $\sim 2.4 R_{\oplus}$.
- Bimodal small-planet population. Defining demographic feature.

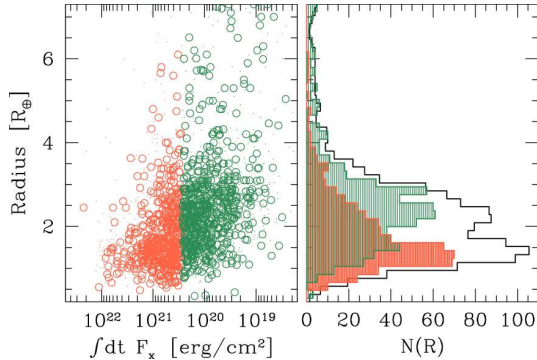
Photoevaporation Strips H/He Envelopes

Energy-limited escape rate:

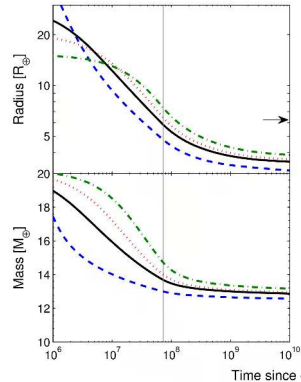
$$\dot{M} \approx \frac{\epsilon \pi F_{\text{XUV}} R_p^3}{GM_p}$$

- Young stars saturate XUV at $\sim 10^3 \times$ present-day for ~ 100 Myr.
- Low-density, low-mass planets are *easier to strip*.
- Cumulative loss for a $10 M_{\oplus}$ sub-Neptune at 0.1 AU: a few % of an Earth mass.
- Comparable to a primordial H/He envelope on a rocky core.
- Owen & Wu (2013): naturally produces a bimodal radius distribution.

Photoevaporation Model + Mass-Loss Tracks

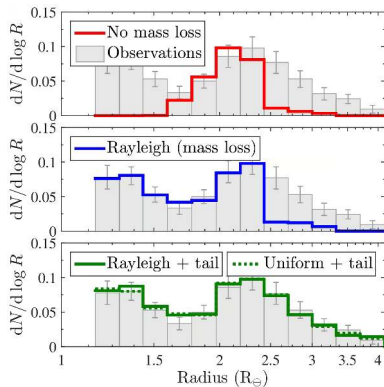


Left: predicted radius valley



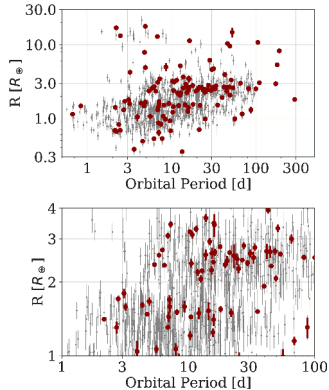
Right: time evolution of radius + mass; saturated XUV phase out to ~ 100 Myr drives bimodal final distribution.

Core-Powered Mass Loss: Same Outcome, Different Driver



- Ginzburg+ 2018, Gupta+ 2019.
- Hot rocky core retains formation heat $\rightarrow$ basal flux into envelope drives slow outflow.
- No XUV needed.
- **Bottom panel** (Rayleigh + tail): bimodal radius distribution at $\sim 1.8 R_{\oplus}$, mirroring photoevaporation.
- Both mechanisms likely operate.

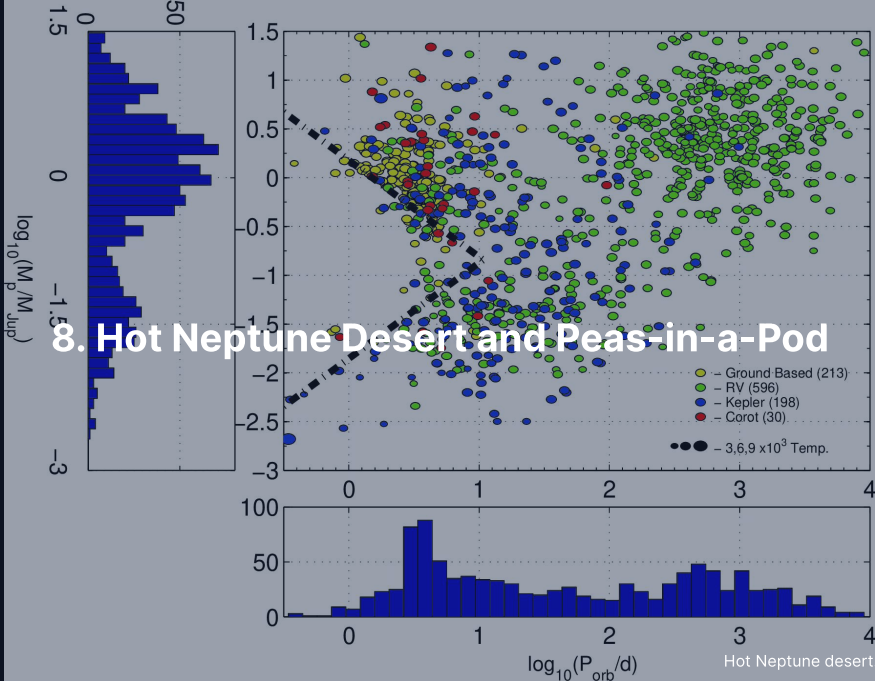
Slope of the Valley with Period



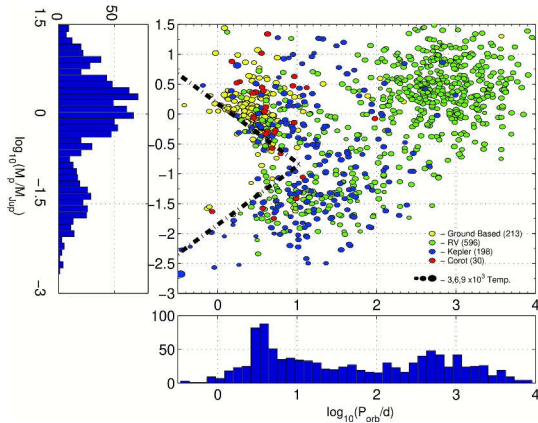
- Van Eylen+ 2018, asteroseismic radii.
- Valley shifts to smaller R_p at longer periods.
- Brown: super-Earths. Blue: sub-Neptunes.
- Slope consistent with both photoevaporation and core-powered models.
- Current data cannot distinguish.

Implication: Many Super-Earths Are Stripped Cores

- Many close-in rocky super-Earths are not “naturally” rocky.
- They are the bare cores of former sub-Neptunes.
- Whether a given super-Earth was born rocky or stripped: *cannot tell from bulk density alone*.
- Consequence for habitability: stripped cores have violent atmospheric histories and may have very different surface/interior chemistry.



The Hot Neptune Desert

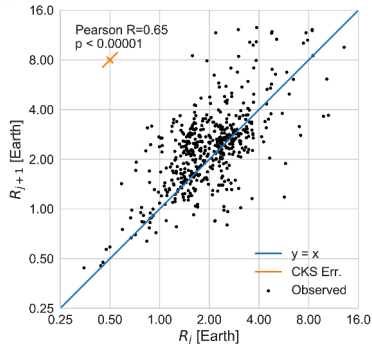


- Striking deficit at $10\text{--}100 M_{\oplus}$, $P < 5$ d.
- Hot Jupiters at the same periods: rare but present.
- Ultra-short rocky: rare but present.
- Hot Neptunes: *essentially absent*.
- Power-law boundaries (Mazeh+ 2016).

Two Edges, Two Mechanisms

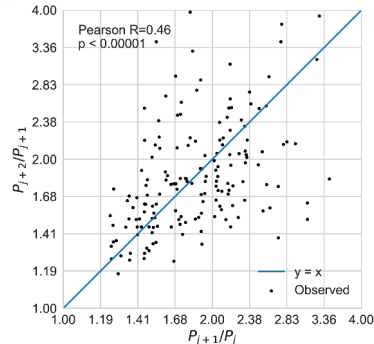
- **Lower edge (small mass):** photoevaporation strips weakly-bound Neptune envelopes → leaves super-Earths.
- **Upper edge (large mass):** Roche-lobe overflow on inflated hot Jupiters truncates the heavy population.
- Independent evidence that mass loss is the dominant sculptor of close-in planets.

Peas-in-a-Pod



Adjacent radii: Pearson 0.65

Within a Kepler multi-system, planets tend to be *the same size* as each other and *uniformly spaced*. Weiss et al. (2018).

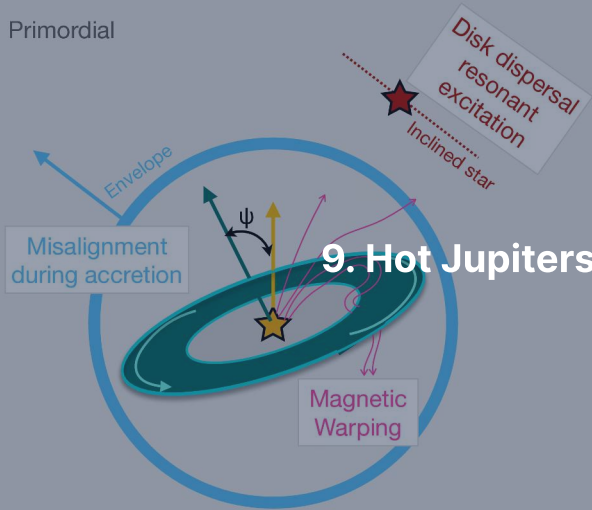


Adjacent period ratios: clustered

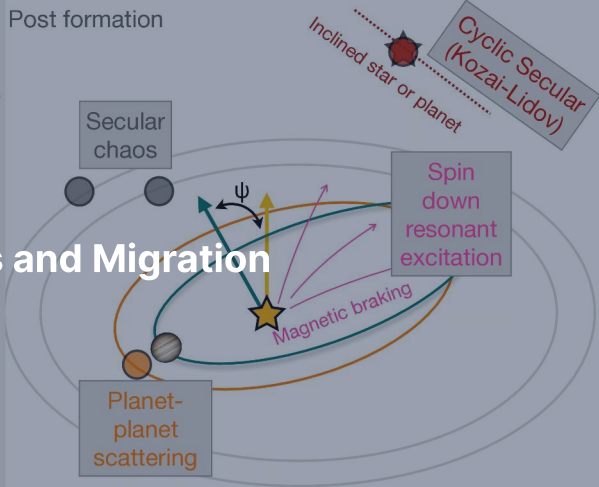
Peas-in-a-Pod: Implications

- Implies smooth, local formation rather than stochastic giant impacts.
- Inner solar system (giant-impact dominated) may be unusual.
- TRAPPIST-1: extreme example of a peas-in-a-pod with resonant chain.
- Kepler-90 (8 planets), TOI-178 (6 planets in unbroken chain).
- Not universal: RV-discovered systems show much more architectural diversity.

Primordial



Post formation



9. Hot Jupiters and Migration

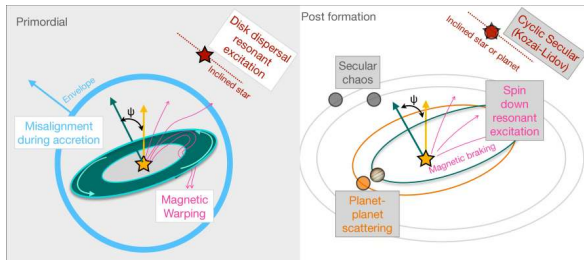
Three Migration Channels

Hot Jupiters at 0.05 AU *must* have migrated.

- **Disk migration** (Type II): smooth, gas-driven. Preserves alignment.
- **High-eccentricity migration** (Kozai-Lidov): distant perturber pumps eccentricity, tides circularise. Produces misalignments.
- **Planet-planet scattering**: dynamical instability ejects giants, leaves survivors on eccentric orbits, tides circularise. Misalignments.

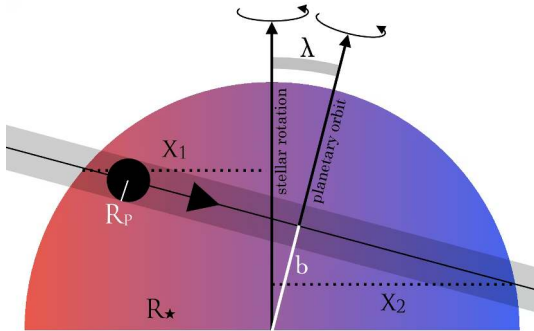
Inner edge set by Roche limit + tidal circularisation timescale: pile-up at $\sim 0.04\text{--}0.05$ AU.

Migration Channels and Stellar Obliquities



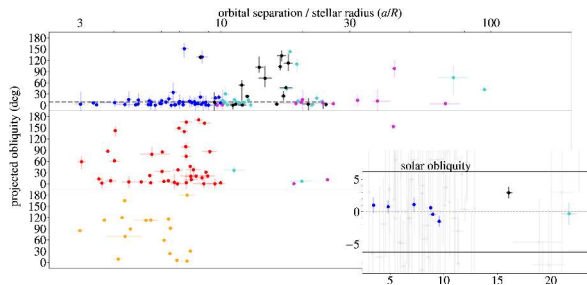
- Albrecht+ 2022.
- Disk migration → aligned.
- Kozai-Lidov → arbitrary obliquity.
- Scattering → misaligned.
- Stellar obliquity is the main empirical handle.

Rossiter-McLaughlin Effect

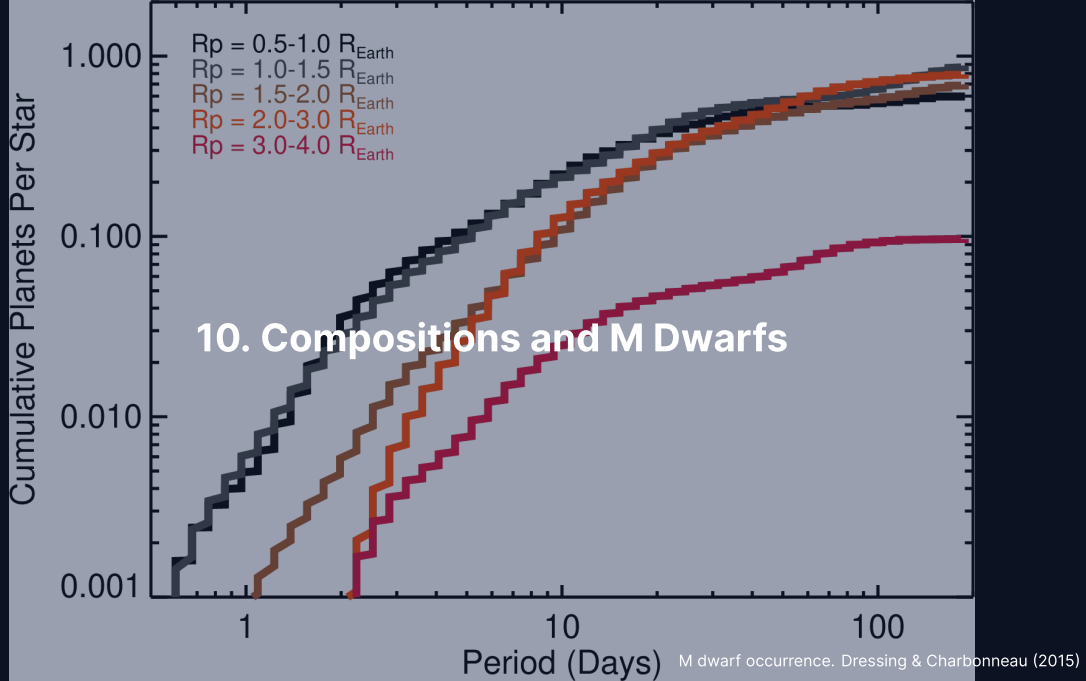


- Transiting planet first occults blueshifted (approaching) hemisphere, then redshifted.
- Time-resolved line distortion traces angle λ between projected stellar spin and orbit normal.
- Triaud (2018) review.

Observed Obliquity Distribution



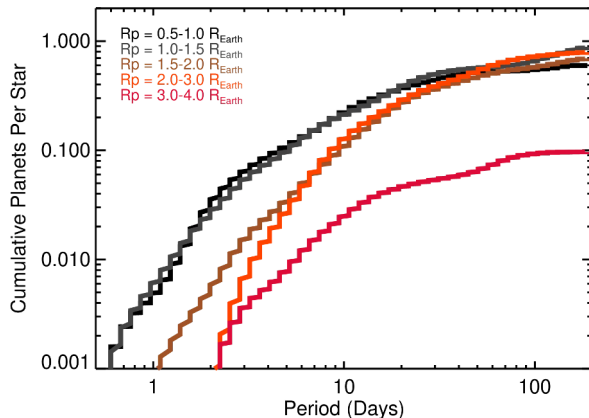
- λ vs a/R_* , with sub-panels per host spectral type (panel labels too small for back rows; trust the trend, not the tick labels).
- Cool stars ($T_{\text{eff}} < 6250$ K): tight orbits aligned (tidal realignment).
- Hot stars: scattered λ .
- Wide scatter at large a/R_* implies a wide *primordial* obliquity distribution.
- All three migration channels likely operate.



Sub-Neptunes: Water World or H/He?

- Super-Earths (below valley): rocky, $\bar{\rho} \sim 4\text{--}8 \text{ g cm}^{-3}$.
- Sub-Neptunes (above valley): $\bar{\rho} \sim 1\text{--}3 \text{ g cm}^{-3}$, need volatiles on top of a rocky core.
- Volatile component degenerate: thick H/He, thick H_2O , or some mix.
- **Hycean** hypothesis (Madhusudhan+ 2023): shallow liquid ocean under thick H_2 atmosphere. Contested.
- Ultra-short-period rocky: thin atmosphere vs bare rock indistinguishable from density alone.

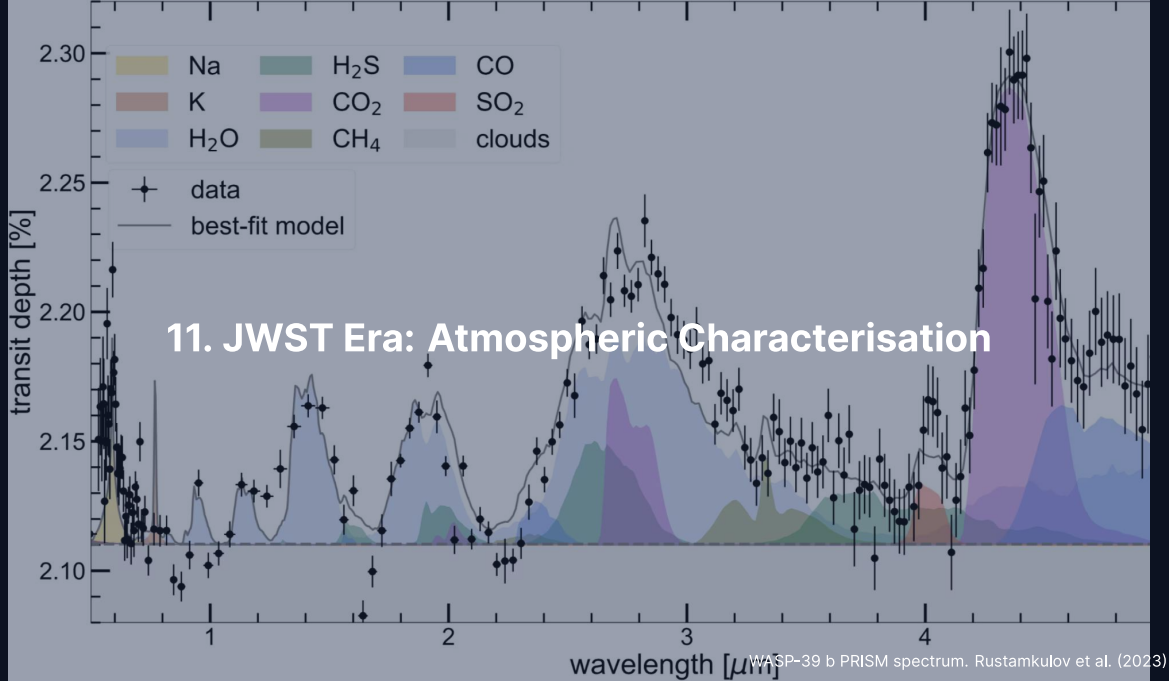
M Dwarfs: Easy Targets, Tough Habitability



- ~ 75% of all main-sequence stars.
- M dwarfs host ~ 2.5 small planets each; ~ 0.16 in HZ.
- Transit + RV signals geometrically favourable.
- BUT: pre-main-sequence high-luminosity for hundreds of Myr.
- Habitable-zone planet *starts* inside runaway greenhouse zone.
- Dressing & Charbonneau (2015).

M Dwarf Habitability: Two Worries

- **Stellar activity** + early high-XUV → Luger & Barnes (2015) showed an ocean's worth of water can be lost as O_2 (false-positive biosignature).
- **Tidal locking**: HZ M dwarf planets in 1:1 spin-orbit resonance.
- Permanent dayside / nightside; climate must redistribute heat or condense volatiles on the night side.
- 3D GCMs (Yang+ 2013): substellar clouds can extend the inner edge.
- TRAPPIST-1 = central laboratory for these questions.



Transmission Spectroscopy

Atmospheric scale height:

$$H = \frac{k_B T}{\mu m_u g}$$

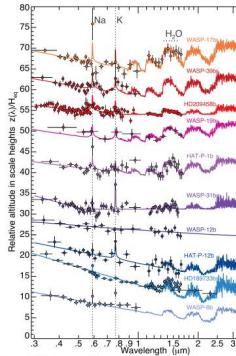
Wavelength-dependent depth: $\delta(\lambda) = (R_p + n_H(\lambda) H)^2 / R_\star^2$.

H is wavelength-independent; $n_H(\lambda)$ is the number of scale heights probed (typically a few).

Quick scales for hot Jupiter ($T = 1500$ K, $\mu = 2.3$, $g = 25$ m s⁻²):

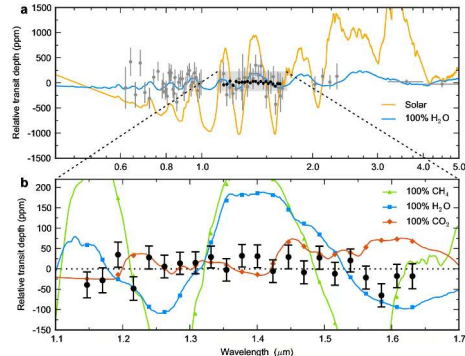
- $H \approx 200$ km
- Modulation across ~ 5 scale heights: $\Delta\delta/\delta \approx 2 \times 10^{-2}$
- Absolute modulation: ~ 200 ppm $\rightarrow$ JWST-routine.
- Sub-Neptune: 10–100 ppm. Earth/M dwarf: <10 ppm.

Clouds Can Mute Spectra

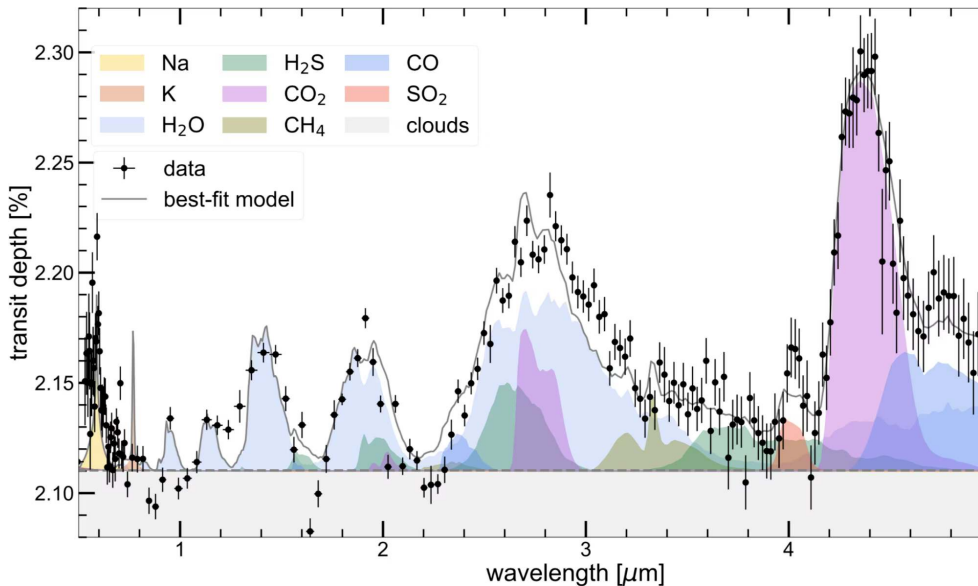


HST + Spitzer ten hot Jupiters

Continuum from clear (Na, K, H₂O) to fully cloudy. Diversity at similar T , g is one of the central puzzles.

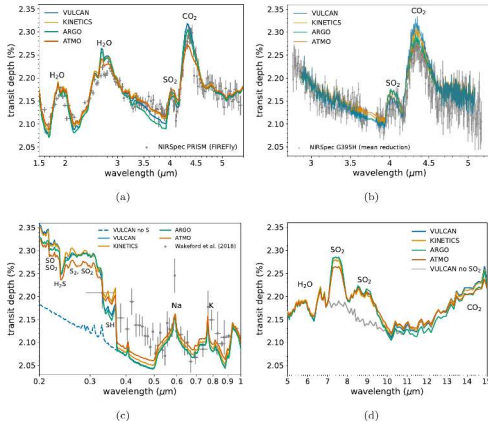


GJ 1214 b: featureless. Kreidberg+ 2014.



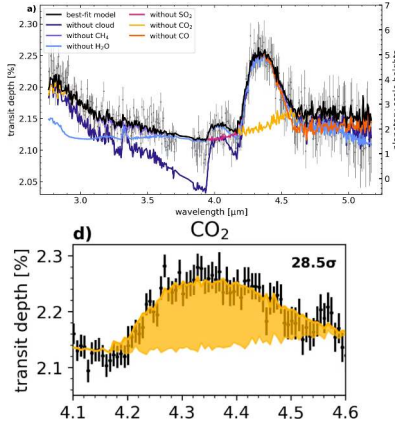
JWST/NIRSpec PRISM transmission spectrum of WASP-39 b: Na, K, H₂O, CO, CO₂, SO₂, CH₄, clouds. Rustamkulov+ 2023.

WASP-39 b: First Disequilibrium Photochemistry



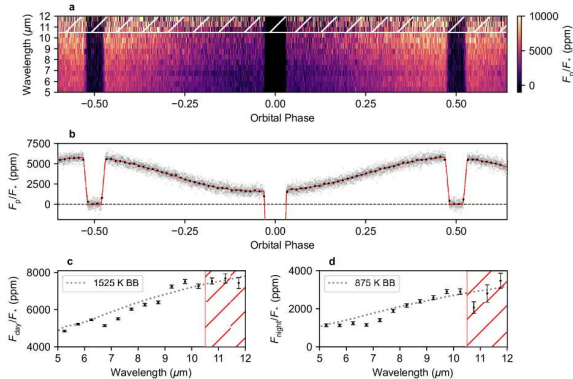
- ERS target: $0.28 M_J$ hot Saturn.
- Clean detection of **photochemical SO_2** .
- Four panels = four photochemistry codes; follow the SO_2 feature at $\sim 4 \mu\text{m}$, not the panel labels.
- SO_2 requires UV photolysis of H_2S ; *not* from equilibrium chemistry.
- Tsai+ 2023. First unambiguous disequilibrium photochemistry.

WASP-39 b: 28.5σ CO₂ Feature



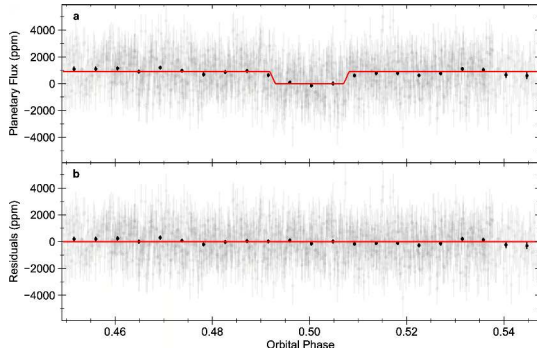
- Per-species decomposition of NIRSpect G395H.
- CO₂ near 4.3 μm : 28.5σ .
- Strongest single absorption feature ever in any exoplanet atmosphere.
- Alderson+ 2023.

WASP-43 b: A Phase Curve in Thermal IR



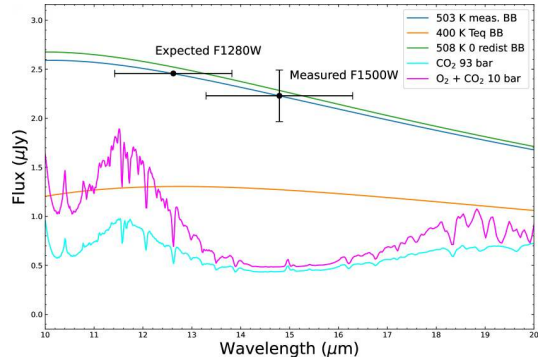
- JWST MIRI phase curve.
- Two transits, two secondary eclipses, smooth phase variation.
- Strong day-night contrast despite thick atmosphere.
- Bell+ 2024.
- Day = hot, night = much colder.

TRAPPIST-1 b: Bare Rock Dayside



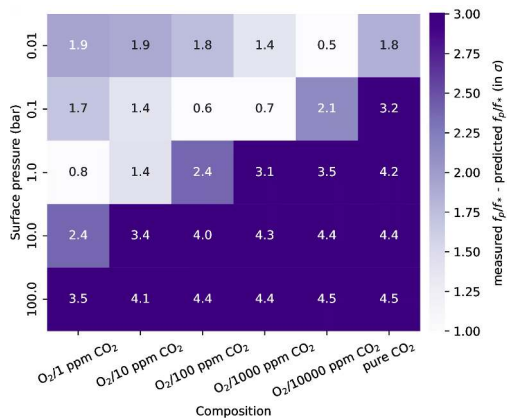
15 μm secondary eclipse

$T_B = 503^{+26}_{-27}$ K, consistent with bare rock equilibrium. **Thick CO₂ atmosphere ruled out.** Greene+ 2023.



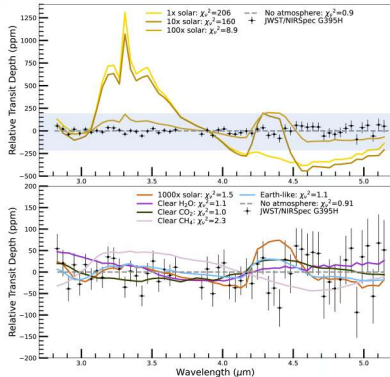
Compared to model atmospheres

TRAPPIST-1 c: Same Story



- Zieba+ 2023.
- Grid of CO₂ partial pressure × thickness.
- ≥ 0.1 bar of CO₂: ruled out.
- Thin atmospheres or none: still possible.
- Inner TRAPPIST-1 planets do not retain Venus-like atmospheres.

LHS 475 b: Featureless

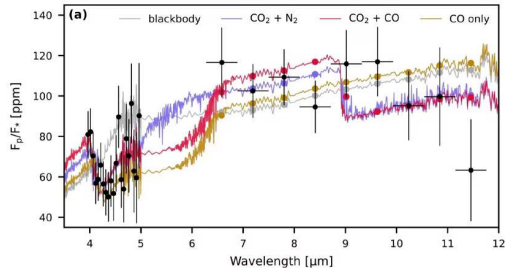


- NIRSpec G395H, Earth-size at 12 pc.
- Flat featureless transmission spectrum.
- H_2/He atmospheres ruled out.
- Pure CH_4 ruled out.
- Pure CO_2 Venus-like marginal.
- Lustig-Yaeger+ 2023.

Pattern: Most Small M Dwarf Planets Lack Atmospheres

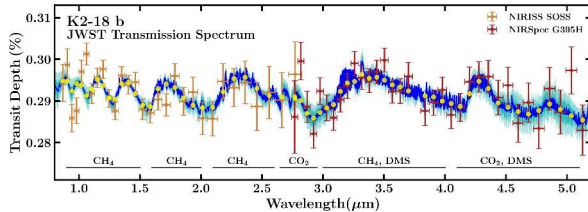
- LHS 475 b, GJ 486 b, GJ 1132 b, TRAPPIST-1 b/c: all consistent with no detectable atmosphere.
- Consistent with Luger & Barnes (2015): early M dwarf XUV strips primordial atmospheres.
- Maybe a watershed: M dwarf HZ planets may not retain water at all.

55 Cancri e: First Atmosphere on a Sun-Like Host?



- Hot rocky super-Earth, ~ 2000 K dayside, plausibly molten.
- NIRCам + MIRI emission.
- Falls below bare-rock blackbody.
- Phase modulation $\rightarrow$ secondary CO/CO_2 atmosphere outgassed from molten surface.
- Hu+ 2024. *First* tentative atmosphere on rocky planet around Sun-like host.

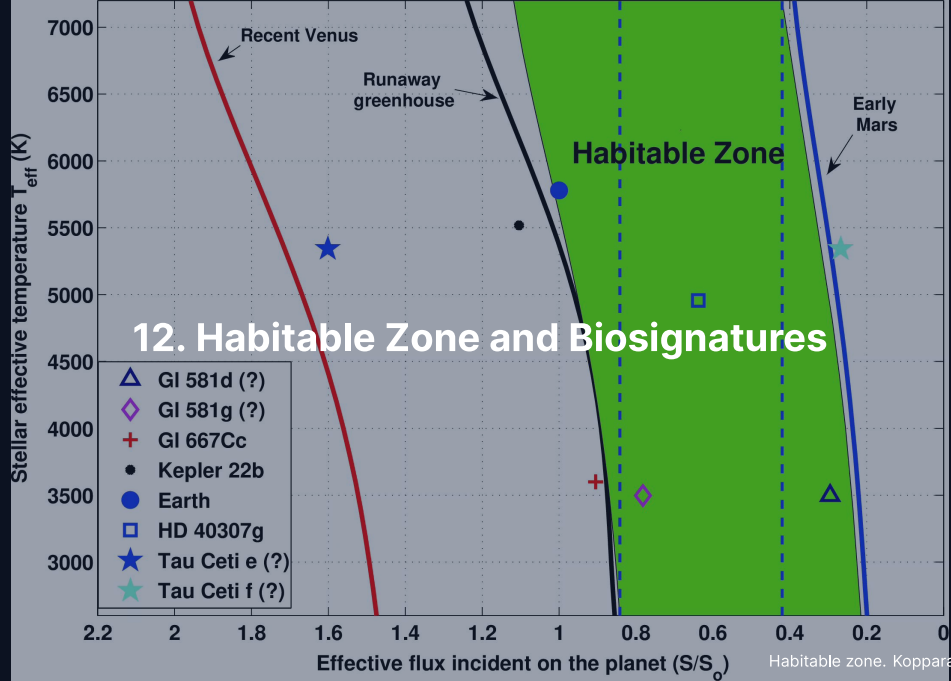
K2-18 b: Biosignature Controversy



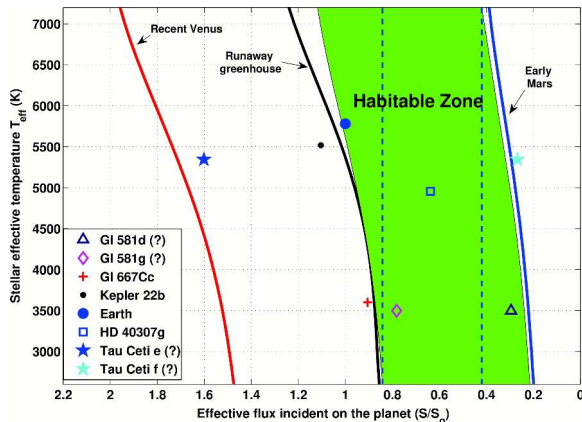
- Sub-Neptune in HZ of M3 host.
- Madhusudhan+ 2023: CH₄ + CO₂ + tentative DMS ($\sim 3\sigma$).
- Interpreted as **hycean**: H₂ atmosphere over ocean.
- Wogan+ 2024: cannot reproduce DMS; mini-Neptune fit.
- Glein+ 2024: geochemistry against habitable ocean.
- Retrieval-dependent. Debate open.

K2-18 b's Pedagogical Value

- Whatever the resolution: K2-18 b is the textbook case of how a tentative biosignature is tested in real time.
- Same pattern: ALH84001, Mars methane, Venus phosphine.
- This time the discussion has been faster, more transparent, better documented.
- Useful template for the louder claims of the next decade.



The Classical Habitable Zone



- Inner edge: Simpson-Nakajima runaway greenhouse (L9).
- Outer edge: maximum CO_2 greenhouse condenses out as ice.
- Kopparapu+ 2013 modernised Kasting+ 1993.
- Symbols: TRAPPIST-1, Tau Ceti, GJ 581/667, etc.

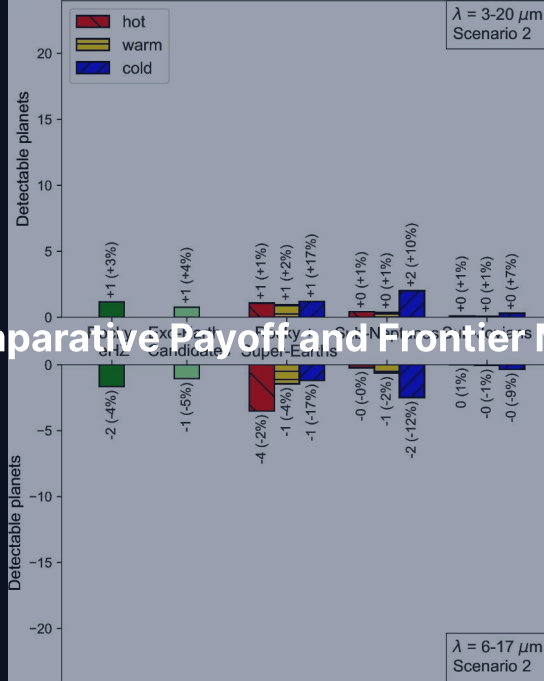
HZ is a Trajectory, Not a Snapshot Line

- History matters more than current location.
- Venus today sits formally inside the HZ if computed naively, yet is hellish (L9).
- M dwarf HZ planets traverse the runaway zone for hundreds of Myr in pre-MS phase.
- 1D Kasting/Kopparapu boundaries shift by 5–20% in 3D GCMs.
- Treat the HZ as a first-order screening tool, not a precise boundary.

Biosignature Gases and False Positives

- Classical: *disequilibrium combinations*. On Earth: $O_2 + CH_4$.
- Catalogue: O_2 , O_3 , CH_4 , N_2O , organosulphur, organohalogen species.
- No single gas in isolation is convincing.
- **False positives are the main challenge.**
 - Wordsworth+ 2014: H_2O photolysis + H escape $\rightarrow$ abiotic O_2 .
 - CO_2 photolysis $\rightarrow$ abiotic O_2 via $CO + O$.
 - Volcanic + serpentinisation $\rightarrow$ abiotic CH_4 .
- Inverse problem: must rule out *all* plausible abiotic pathways.

13. Comparative Payoff and Frontier Missions



Is the Solar System Typical?

Honest answer: **not obviously, but not yet known to be rare.**

Sparsely populated corners:

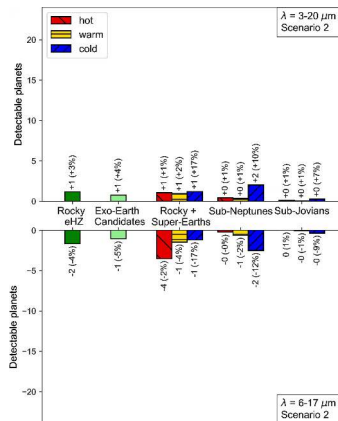
- Most common host: M dwarf, not G dwarf.
- Most common planet: sub-Neptune ($2\text{--}3 R_{\oplus}$). *Solar system has none.*
- Most common architecture: peas-in-a-pod, $\sim 5\text{--}8$ similar planets within 0.2 AU. *Not the inner solar system.*
- No hot Jupiter or hot Neptune in our system.
- Wide low-eccentricity giants (vs RV-discovered eccentric giants).

But: detection biases penalise solar-system analogues. Gaia DR4/DR5 + PLATO will resolve.

Frontier Missions 2026–2035: Transits and Atmospheres

- **PLATO** (ESA, end 2026): Earth analogues in HZ of bright Sun-like stars; asteroseismic stellar radii.
- **Ariel** (ESA, 2029): atmospheric statistics of ~ 1000 exoplanets at 0.5–7.8 μm .
- **Roman** (NASA, 2027): 1000–3000 microlensing planets at 1–10 AU + comparable rogue population; coronagraph tech demo at 10^{-8} – 10^{-9} .

Frontier Missions 2030s–2040s: Direct Imaging



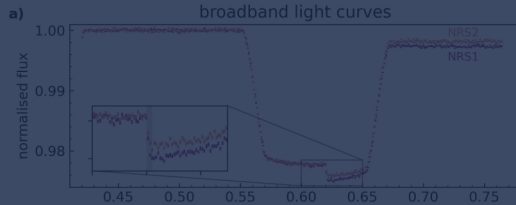
- **HWO** (NASA, 2040s): 6 m, 10^{-10} contrast, ~ 25 Earth analogues.
- **LIFE** (ESA concept): nulling interferometer, 4–18 μm thermal emission.
- Figure: yield-sensitivity panels from Quanz+ 2022; trends matter, not the small numerical labels.
- **ELT class** (ground, 2028+): 39, 24.5, 30 m. High-contrast + high-resolution NIR.

From Demography to Individual Characterisation

- Today: thousands of detected planets, dozens of well-characterised atmospheres.
- 2040s: full atmospheric retrievals on a curated sample of Earth analogues.
- Open question for L14: what would convince us we have *detected life*?
 - Single biosignature gas?
 - Specific combinations at specific abundances?
 - Temporal variability (seasonal cycles)?
 - Vegetation or photosynthetic surface features?

Summary: Today's Course-Level Takeaways

1. Each detection method picks out a different region of parameter space; the observed archive is the union of biases.
2. Transit + RV $\rightarrow$ bulk density: the observational machinery that turned exoplanet detections into physical objects.
3. Kepler showed planets are common. $\eta_{\oplus} \sim 0.3$. Hot Jupiters are rare.
4. **Radius valley** at $\sim 1.8 R_{\oplus}$: photoevaporation + core-powered mass loss as universal sculptor.
5. Hot Neptune desert + peas-in-a-pod + TRAPPIST-1 chain: three architectural results that any formation theory must explain.
6. JWST: WASP-39 b SO_2 , TRAPPIST-1 b/c bare rock, K2-18 b biosignature controversy.
7. Solar system is not obviously typical. Gaia DR4/5 + PLATO will tell us if it is rare.



Questions?

Next: **Lecture 14. Synthesis & Astrobiology**

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience

